Acute Care of Esophageal Cancer

Esophageal Cancer Overview

Triage questions:

- Has EGD been done? Pathology results?
- Presence of metastatic disease on CT chest/abdomen/pelvis?
- Upper GI bleeding requiring transfusion? Is bleeding persistent?
- Can patient maintain hydration orally?

Triage

- Bleeding
- Obstruction (unable to maintain hydration)
- Perforation

Patients who can maintain their hydration orally:

- Outpatient management
- No need for transfer to CMC Main

Patients with dysphagia who can maintain oral hydration

Clinically staged as T3 and fit into one of two caregories:

- Patients with metastatic disease (liver, lung, or extra-regional lymphadenopathy with cervical or para-aortic nodes). Treatment is chemotherapy with radiation for palliation of dysphagia or treatment of symptomatic bone metastasis
- Patients without metastatic disease are staged *Locally Advanced* and treated with preoperative chemotherapy ± radiation followed by surgery.

Surgery as primary therapy is not indicated except for cancer perforation.

Patients with minimal dysphagia

Typically present acutely with bleeding, often while under anticoagulation.

Some of these patients are AJCC T2 and would be candidates for primary surgery.

Most are AJCC T3 and should get preoperative chemo ± radiation.

Endoscopic Ultrasound (EUS) is necessary to distinguish T2 from T3

- Available only at CMC Main
- Usually can be done as an outpatient

500+ esophagectomies over 19 years: None performed emergently for bleeding

Endoscopy

Endoscopy with biopsy is foundational for patients with esophageal obstruction.

Majority will have evidence of malignancy on EGD. Negative endoscopic biopsies in the patient with an obstructing esophageal stricture are unusual but not rare.

Options for re-biopsy include:

- Neonatal (5mm) endoscope
- Dilation followed by biopsy
- Endoscopic ultrasound with FNA.

These options are available at CMC Main and may not be available at regional facilities. (Faigel et al. 1998)

CT Scan

Patients with esophageal cancer need CT chest/abdomen/pelvis with IV contrast for initial staging.

PET/CT Scan

- Needed for complete staging
- Rarely needed in the acute setting
- Required for radiation therapy planning

Prep:

- NPO for 4 hours (except water)
- No IV dextrose for 4 hours prior
- Blood sugars controlled

Endoscopic ultrasound (EUS)

Distinguishes localized tumors (AJCC T2) who may be candidates for primary surgery from patients with locally-advanced cancers (AJCC T3) who need preoperative chemotherapy ± radiation.

Patients with T2 tumors generally have minimal dysphagia and frequently present with bleeding during anticoagulation or surveillance for Barrett's esophagus.

EUS has limited utility in the staging of patients with esophageal cancer who present with dysphagia (Mansfield et al. 2017) (Findlay et al. 2015) (Radlinski et al. 2020) (Ripley et al. 2016) and is not necessary in these patients.

Esophageal Obstruction (can't maintain hydration)

Enteral Feeding Tube

Metastatic disease: PEG

- Most patients can be treated with a neonatal scope + 20Fr Pull PEG

No metastatic disease

- Lap gastrostomy preferred
- Lap jejunostomy never the wrong answer

Laparoscopic Gastrostomy

Laparoscopic gastrostomy is the preferred method for enteral access in patients with esophageal cancer for surgeons familiar with esophageal reconstruction. The stomach is the preferred method of reconstruction during esophagectomy and it is critical that gastrostomy placement not complicate future surgery. Gastrostomy tubes are more convenient for the patient as feedings can be done by bolus feedings, while jejunostomy feeding requires a pump over extended periods (12-16 hours).

Laparoscopic Gastrostomy

The laparoscopic approach allows placement of the tube away from the right gastroepiploic artery. Findings the ideal location for the tube involves laparoscopy with low insufflation pressure (4mm Hg), which allows adequate distention of the stomach with an endoscope. A neonatal (5mm) endoscope can be helpful. A site is selected at the midpoint of the stomach on a saggital plane to allow minimal tension on the tube. The tube is placed as far proximally on the stomach as possible yet still at least 3cm from the left costal margin (for patient comfort and to allow minimal tension on the gastrostomy tube). Operative Description of Laparoscopic Gastrostomy

Laparoscopic Jejunostomy

Jejunostomy is the traditional method of nutritional support for patients with esophageal obstruction. This avoids the risk of injury to the future gastric conduit at the time of esophagectomy.

Description of operative setup for laparoscoic jejunostomy

Postoperative Care for Jejunostomy

Endoscopic stents

Endoscopic stents to treat esophageal obstruction are appropriate in patients with documented metastatic disease (Stage IV) and those who will never be candidates for surgery.

In patients who are potentially candidates for esophagectomy, stents are generally avoided to prevent complications of surgery through fibrosis of the esophagus proximal to the tumor.

Stent vs Gastrostomy Feeding Tube

A retrospective study of patients with esophageal cancer and dysphagia (Min et al. 2017) found that treatment with gastrostomy was associated with longer survival than those treated with a endoscopic stent and had higher albumin levels and less need for re-intervention.

Percutaneous Endoscopic Gastrostomy

For patients with Stage IV disease, percutaneous endoscopic gastrostomy (PEG) is indicated for nutritional support. A gastrostomy allows for bolus feeding and is more convenient for patients than a jejunostomy, which requires an infusion pump.

For patients with potentially resectable disease, PEG is avoided to prevent injury to the right gastroepiploic artery and the risk of tumor seeding of the gastrostomy tract.

Central Venous Port

Central venous port can facilitate the administration of chemotherapy but is not essential in the acute management of patients with esophageal cancer, particularly patients with poor performance status who may not receive chemotherapy once they meet with their treating medical oncologist.

Inpatient medical oncology consultation can help sort out which patients are not candidates for chemotherapy.

Perforated Esophageal Cancer

Triage questions:

- Has EGD been done? Pathology results?
- Spontaneous or s/p procedure?
- Has patient had prior radiation therapy?
- CT Chest/abdomen/pelvis
- Pleural effusion or pneumothorax?
- Pneumoperitoneum or peritoneal fluid?

Patients with perforated esophageal cancer need multidisciplinary emergency management. These patients should be transferred emergently to CMC Main.

Patients with pleural effusion need emergent thoracic surgery consultation. Patients with pneumoperitoneum or intra-abdominal fluid in the upper abdomen need emergent laparoscopic placement of drains.

Triage Team

- CMC Thoracic Surg Attending SHVI (for thoracic perforations)
- Red LCI SO (for abdominal perforations)
- CMC LCI Head and Neck Surg Onc Attending (for cervical perforations)

Acute Management

- IV hydration
- Correction of electrolyte abnormalities
- IV antibiotics
- CT Esophagram with water soluble contrast
- CT abdomen/pelvis
- CT neck (if suspicion of cervical perforation)

Inpatient Management

- Emergent drainage of pleural effusions
- Emergent laparoscopic placement of drains for evidence of intra-abdominal perforation

Background

Perforated esophageal cancer most commonly occurs after endoscopy, endoscopic ultrasound, or dilation of an esophageal cancer (or benign stricture). Endoscopic stenting is frequently not feasible due to the underlying stricture. Treatment options are conservative management with drainage and IV antibiotics, endoluminal stenting, or emergency surgery. (Abu-Daff et al. 2016)

Diagnosis

CT esophagram with water soluble contrast is the diagnostic study of choice(Suarez-Poveda et al. 2014). This should include CT neck and/or CT abdomen/pelvis depending upon the suspected level of perforation.

A study from Colorado (Madsen et al. 2023) of 65 patients transferred with suspected esophageal perforation found that 24 patients were ultimately found not to have a perforation. Among patients for whom pneumomediastinum was their sole criteria for transfer, only 57% were ultimately found to have a perforation. Among patients with a pleural effusion prior to transfer, 83% were ultimately found to have a perforation. Patients with a lactate greater than 1.6 were 11 times more likely to have aperforation.

Cervical esophageal perforation

Contained perforations of the cervical esophagus can be treated with NPO and IV antibiotics. Uncontained perforations need surgical exploration. In this case, nutritional support with gastrostomy is indicated.

Thoracic esophageal perforations

Pittsburgh esophageal perforation scoring system: (Abbas et al. 2009) Validation of Pittsburgh scoring system (Schweigert et al. 2016)

A series from Birmingham UK of 74 esophageal perforations over 15 years include 13 patients with perforated cancers, of whom 7 were treated with endoscopic stenting (**charalampakis1710?**). These 13 patients had a 1-year mortality of 77%

Conservative management

There is little experience in the literature for the conservative management of perforated esophageal cancer. Most clinical reports exclude patients with perforated cancers. A report from Kenya of 10 perforations among 492 dilations of esophageal cancer, of whom 9 were successfully treated with endoluminal stents (White et al. 2003). Stenting of perforated esophageal cancer is limited to case reports (Kobayashi et al. 2019)

Emergent Esophagectomy

Emergency esophagectomy is the traditional approach to perforated esophageal cancer given the difficulty in stenting an area of tumor (or stricture). Short-term goals are fluid resuscitation, IV antibiotics, and drainage of the pleurae (and abdomen).

Emergent esophagogastrectomy via an abdominal and thoracic approach is the conventional approach in a fit patient. For frail patients, an transabdominal approach with esophagogastrostomy is less morbid (Gillen et al. 2009) although placing the patient at higher risk of postoperative reflux.

Orientation Manual

References

Abbas, Ghulam, Matthew J. Schuchert, Brian L. Pettiford, et al. 2009. "Contemporaneous Management of Esophageal Perforation." *Surgery* 146 (4): 749-755; discussion 755-756. https://doi.org/10.1016/j.surg.2009.06.058.

Abu-Daff, S., F. Shamji, J. Ivanovic, et al. 2016. "Esophagectomy in Esophageal Perforations: An Analysis." *Diseases of the Esophagus: Official Journal of the International Society for Diseases of the Esophagus* 29 (1): 34–40. https://doi.org/10.1111/dote.12294.

Faigel, D. O., C. Deveney, D. Phillips, and M. B. Fennerty. 1998. "Biopsy-Negative Malignant Esophageal Stricture: Diagnosis by Endoscopic Ultrasound." *The American Journal of Gastroenterology* 93 (11): 2257–60. https://doi.org/10.1111/j.1572-0241.1998.00629.x.

Findlay, J. M., K. M. Bradley, E. J. Maile, et al. 2015. "Pragmatic Staging of Oesophageal Cancer Using Decision Theory Involving Selective Endoscopic Ultrasonography, PET and Laparoscopy." *The British Journal of Surgery* 102 (12): 1488–99. https://doi.org/10.1002/bjs.9905.

Gillen, Sonja, Helmut Friess, and Jörg Kleeff. 2009. "Palliative Cardia Resection with Gastroesophageal Reconstruction for Perforated Carcinoma of the Gastroesophageal Junction." *World Journal of Gastroenterology : WJG* 15 (24): 3065–67. https://doi.org/10.3748/wjg.15.3065.

Kobayashi, Teruyuki, Tomoki Makino, Makoto Yamasaki, Koji Tanaka, Masaki Mori, and Yuichiro Doki. 2019. "Successful Stenting Followed by Surgery for Perforated Esophageal Cancer Due to Chemotherapy." *The Annals of Thoracic Surgery* 108 (6): e361–63. https://doi.org/10.1016/j.athoracsur.2019.04.022.

Madsen, Helen J., Christina M. Stuart, Brandon M. Wojcik, et al. 2023. "Esophagram Should Be Performed to Diagnose Esophageal Perforation Before Inter-Hospital Transfer." *Journal of Thoracic Disease* 15 (6): 2984–96. https://doi.org/10.21037/jtd-22-1798.

Mansfield, Sara A., Samer El-Dika, Somashekar G. Krishna, Kyle A. Perry, and Jon P. Walker. 2017. "Routine Staging with Endoscopic Ultrasound in Patients with Obstructing Esophageal Cancer and Dysphagia Rarely Impacts Treatment Decisions." *Surgical Endoscopy* 31 (8): 3227–33. https://doi.org/10.1007/s00464-016-5351-6.

Min, Yang Won, Eun Young Jang, Ji Hey Jung, et al. 2017. "Comparison Between Gastrostomy Feeding and Self-Expandable Metal Stent Insertion for Patients with Esophageal Cancer and Dysphagia." *PloS One* 12 (6): e0179522. https://doi.org/10.1371/journal.pone.0179522.

Radlinski, Mark, Linda W. Martin, Dustin M. Walters, et al. 2020. "Use of Endoscopic Ultrasound in Pre-Treatment Staging of Esophageal Cancer Did Not Alter Management Plan." *Journal of Thoracic Disease* 12 (10): 5850–56. https://doi.org/10.21037/jtd-20-1299.

Ripley, R. Taylor, Inderpal S. Sarkaria, Rachel Grosser, et al. 2016. "Pretreatment Dysphagia in Esophageal Cancer Patients May Eliminate the Need for Staging by Endoscopic Ultrasonography." *The Annals of Thoracic Surgery* 101 (1): 226–30. https://doi.org/10.1016/j.athoracsur.2015.06.062.

Schweigert, Michael, Hugo Santos Sousa, Norbert Solymosi, et al. 2016. "Spotlight on Esophageal Perforation: A Multinational Study Using the Pittsburgh Esophageal Perforation Severity Scoring System." *The Journal of Thoracic and Cardiovascular Surgery* 151 (4): 1002–9. https://doi.org/10.1016/j.jtcvs.2015.11.055.

Suarez-Poveda, Tatiana, Carlos H. Morales-Uribe, Alvaro Sanabria, et al. 2014. "Diagnostic Performance of CT Esophagography in Patients with Suspected Esophageal Rupture." *Emergency Radiology* 21 (5): 505–10. https://doi.org/10.1007/s10140-014-1222-4.

White, Russell E., Caesar Mungatana, and Mark Topazian. 2003. "Expandable Stents for Iatrogenic Perforation of Esophageal Malignancies." *Journal of Gastrointestinal Surgery: Official Journal of the Society for Surgery of the Alimentary Tract* 7 (6): 715-719; discussion 719-720. https://doi.org/10.1016/s1091-255x(03)00064-7.